

CAN VISION-LANGUAGE MODELS LEARN TO EXECUTE CLASSICAL SEARCH ALGORITHMS?

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ABSTRACT

We train vision-language models to execute declared classical search algorithms step by step against a no-repair runtime and test whether such training can be measured to improve search choice. On the additive best-first cells of the original enumeration contract, where an expansion must submit its complete candidate set and heap order ignores submission order, a random-valid control is decision- and expansion-identical to the exact classical reference on 48/48 pairs, so choice quality is unmeasurable there by construction. The BFS and width-based best-first cells of the same contract do register choice (random-valid 15/24 and 17/24 versus exact 24/24), and the learned policy scores 0/24 on each. Menu permutation leaves 36/36 successes with 0/512 invalid operations, which is uninformative about order under submission-order invariance. Injecting schema-valid but inapplicable distractors collapses success to 0/36 at a distractor pick-rate of 0.6, against a chance rate of 0.592 to 0.625, so the policy does not reject such entries. We therefore redesign the decision as a choice-frontier contract, in which the policy selects which runtime-owned frontier state to expand under a binding decision budget. Its pre-registered identity gate is a sanity check that holds by construction whenever a menu offers two or more states. It passes on 9/9 tasks (18 cells, control sequences coincide across the two algorithms), and random-valid and exact separate (random-valid minus exact reference -0.644 , 95% CI $[-0.856, -0.422]$).

In the first measurement the adapter learns to operate the contract, solving 7/18 cells with zero invalid operations ($+0.389$ [$+0.167, +0.611$] over the pretrained base). Choice quality beyond the named references is not established. The contrast against random-valid is $+0.033$ [$-0.033, +0.133$] and rests on a single task. An exploratory comparator zoo finds no positive learned Pareto dominance, and the adapter’s solve-versus-budget score ($.1042$) lies between `bfs-order` ($.0694$) and novelty-first selection ($.1250$) (point estimates, intervals overlap). This is not evidence of equivalence, since no equivalence margin was pre-registered. The single positive learned contrast is on a 15-task width-based best-first development panel of the enumeration contract (0.933 versus 0.467 for random-valid). All results are development-stage and single-seed except two replication cells, saturating controls bound measurability rather than policy quality, and the held-out final evaluation has not been run. The paper reports a measurement result, not planning ability, together with two cheap benchmark-hygiene checks, the per-decision identity test and menu manipulation.

1 INTRODUCTION

On classical planning probes (Valmeekam et al., 2023b; Stein et al., 2025; Kokel et al., 2026), off-the-shelf language models improve at operational reasoning with scale and prompting but not at structural enumeration (Huang et al., 2026), leaving open whether a model can be trained to sustain a declared search procedure over a budgeted episode. We train a Search Process Policy (Section 3) to emit Typed Search Operations that a no-repair Trusted Search Runtime validates against Algorithm Invariants, with every exploration-determining operand emitted so that replay reproduces the trace, and we measure validity and success separately from advantage over an oracle-assisted random-valid control. The algorithms are BFS, complete unpruned width-based best-first search (BFWS), and weighted A* (weight 3) and greedy best-first over h_{add} , under text, visual, and

multimodal observations (Section 4). All cells are single-seed except two replication cells (seeds 17/29/71, descriptive $n=3$), and the held-out final primary evaluation has not been executed, so no final comparative efficacy finding exists.

Under the original enumeration contract (Section 5), a per-decision identity audit finds random-valid and exact reference decision- and expansion-identical on 48/48 pairs of the additive best-first cells, because an expansion submits its complete candidate set and heap order ignores submission order, so choice quality is unmeasurable there and saturating random-valid entries measure operation validity, while the BFS and width-based cells register choice and distractor injection shows the policy does not reject inapplicable menu entries. One positive learned contrast exists, on a 15-task text-state BFWS development panel (one fixed-seed run 0.933, random-valid 0.467, base 0.0, exact 1.0). On the 24-task baseline, process SFT solved 125 of 288 versus 240 for random-valid, 0 for the base, and 288 for exact, all in additive cells with no decision to make. Over 2,910 replayed episodes learned failures are early invariant rejections of applicable operations, while random-valid fails only late on budget exhaustion. Text shuffling costs -0.944 (text) and -0.722 (multimodal) against at most -0.111 for visual corruption. An InternVL3.5-8B greedy cell reaches 31/36 (base 0/36, random-valid 24/24 per modality), replicating contract fluency, with its earlier 0/36 BFS result consistent with a choice-requirement difference rather than a backbone null. DAgger added no advantage over exposure-matched continued SFT (1/72 versus 1/72), model-generated successors dropped goal success from 45/45 to 1/45, and no transfer comparison survived Holm correction (0/36).

The redesign (Figure 2) makes the policy select which runtime-owned frontier state to expand under a budget of $2\times$ the additive exact-reference expansion count. Its pre-registered identity audit, a sanity check that holds by construction whenever a menu offers two or more states, passes on 9/9 tasks (18 cells, control sequences coincide across the two algorithms), and random-valid and exact separate (-0.644 , frozen 95% CI $[-0.856, -0.422]$, post-hoc task-clustered $[-0.889, -0.333]$). The adapter solves 7/18 cells with zero invalid operations, a contract-fluency gain of $+0.389$ [$+0.167, +0.611$] over the base, but its contrast against random-valid is $+0.033$ $[-0.033, +0.133]$, not material, with policy-level resolution resting on one discriminative task. An exploratory comparator zoo finds no positive learned Pareto dominance under a fifteen-test Holm family. On solve-versus-budget AUC the adapter (.1042) lies between `bfs-order` (.0694) and novelty-first selection (.1250) with overlapping intervals, random-valid (.1389) is the floor reference rather than a selector, and the selectors read privileged information the model does not see. With no pre-registered margin, non-significance is neither equivalence nor a generalized negative capability claim.

This paper makes three contributions.

- **A measurability audit for grounded-menu search evaluation.** Submission-order invariance makes random-valid decision-identical to the reference on the additive cells of the enumeration contract (48/48 pairs), while its BFS and width-based cells register choice. The audit is two cheap checks, a per-decision identity test and menu manipulation by permutation and distractor injection.
- **An audit-driven choice-frontier contract.** On a replay-determinative, no-repair policy/runtime interface, the policy chooses which frontier state to expand under a binding budget. Its pre-registered identity gate (a by-construction sanity check) passes on 9/9 tasks and random-valid and exact separate, with policy-level resolution resting on one discriminative task.
- **A development-stage map with its negative boundaries.** The choice-frontier measurement shows contract fluency without demonstrated choice quality beyond the named references or positive learned Pareto dominance, and the enumeration contract has one positive contrast on the 15-task width-based panel. Failures localize to invariant rejections, text shuffling outweighs visual corruption, DAgger adds no advantage, model-generated successors collapse goal success, and transfer leaves no comparison surviving correction, all development-stage rather than final efficacy.

The paper therefore reports a measurement result and a reusable audit rather than planning ability. The held-out final primary evaluation has not been executed, and the full-scope v1 generalization and second-backbone branches were withdrawn after their terminal VALID_STOP receipts (closed 2026-09-21) and will not be executed.

2 RELATED WORK

Planning benchmarks (Valmeekam et al., 2023a;b; Stein et al., 2025; Kokel et al., 2026) score plans or plan judgments rather than step-by-step execution of a declared algorithm such as A* (Hart et al., 1968) or BFWS (Lipovetzky & Geffner, 2017), and trace-trained models (Lehnert et al., 2024; Gandhi et al., 2024) and step imitators (Veličković et al., 2020; Zaremba & Sutskever, 2014) lack typed operations and in-episode validation.

In Tree of Thoughts (Yao et al., 2023) and RAP (Hao et al., 2023) a scaffold or UCT policy owns exploration, whereas our model issues the operations that determine it, and Scratchpads (Nye et al., 2021), per-step labels (Lightman et al., 2024), Visual Sketchpad (Hu et al., 2024), and MVoT (Li et al., 2025) supervise or generate unchecked free-form steps and imagery rather than attributing control to an observation channel. Unlike repair loops (Kambhampati et al., 2024; Madaan et al., 2023) and syntax-only constrained decoding (Scholak et al., 2021; Willard & Louf, 2023), our runtime never repairs and checks algorithmic invariants on every operation.

Evaluation-validity work in planning and reasoning warns that common-sense tasks can confound planning with retrieval (Valmeekam et al., 2023a), and embodied suites such as BEHAVIOR-1K broaden task coverage (Li et al., 2024). Research on multimodal training documents VQA models relying on language priors rather than image content, video-language evaluation reports selection bias toward answer positions (Zhu et al., 2020; Loginova et al., 2025), and LVLM chain-of-thought audits find that visual or format cues can influence answers without appearing in reasoning traces (Balasubramanian et al., 2025). Process supervision supplies feedback on intermediate reasoning steps (Lightman et al., 2024), but that training signal alone does not test whether a grounded action menu makes frontier choices identifiable. Prior text-game work treats access to valid actions as an evaluation handicap. Jericho lists world-change detection for identifying valid actions among its optional handicaps, and CALM reports that on half of its unseen games it is competitive with or better than models with access to ground-truth admissible actions (Hausknecht et al., 2020; Yao et al., 2020). The menu manipulations in this paper target this interface by testing whether a trained search policy follows menu surface rather than grounding its operation in state. This paper operationalizes the known admissible-action handicap in search-execution evaluation with a per-decision identity audit of random-valid against the exact reference on the additive best-first cells of the grounded-menu contract, then assesses the redesigned frontier selector against named algorithms rather than treating random-valid as a capability ceiling (Section 5).

3 SEARCH PROCESS POLICY

An episode fixes (x, A, m, B) : a task $x = (D, s_0, G)$ with deterministic grounded domain D , a declared algorithm A given as input, a modality adapter m , and a frozen budget B . At step t the policy sees $o_t = \phi_m(c_t)$, a text-state, visual-state, or multimodal-state rendering of the decision context that must be decision-sufficient (exposing every fact the exact teacher uses at t), with Search Memory and candidates kept textual.

Typed Search Operation. The policy emits $u_t = (\tau_t, \theta_t)$, a type from A ’s vocabulary plus every exploration-determining operand, and an emission that omits one is invalid. **Search Memory.** M_t is bounded runtime-held state (frontier, visited table, BFWS novelty counters), not model state, and o_t is the policy’s only channel to history. **Algorithm Invariant.** Deterministic per-algorithm properties are checked on every operation, namely FIFO order (BFS), novelty and goal-count priority (BFWS (Lipovetzky & Geffner, 2017)), and $g + 3h_{\text{add}}$ or h_{add} order under a declared tie-break (weighted A* (Hart et al., 1968; Pohl, 1970) and greedy best-first over h_{add} (Bonet & Geffner, 2001)), and compliance establishes neither termination, completeness, optimality, nor success. **Trusted Search Runtime.** The runtime only validates, applies, persists, or rejects, never choosing omitted operands, reordering candidates, or repairing, so raw emissions replay to the same trace (schemas and invariants in Appendix A).

Process supervision is token-level cross-entropy cloning of the teacher’s operation on Search-Trace Segments, bounded slices of replay-verified exact traces collected in a dataset $\mathcal{D}$, which optional DAGger (Ross et al., 2011) extends with teacher corrections on visited contexts.

4 EXPERIMENTAL DESIGN

Each algorithm-by-modality-by-condition cell asks whether a trained Search Process Policy exceeds an oracle-assisted validity control on full-episode budgeted success (Section 3), under frozen protocols, a 336 GPU-hour cap, and prespecified gate receipts. Every executed stage is development-stage evidence. Panels are frozen and outcome-blind, the held-out manifest has never been accessed, and every learned cell uses one training seed (17) except two replicated cells (seeds 29 and 71 added, protocol frozen 2026-09-20).

Each cell holds the pretrained base, process SFT, and process SFT plus two DAgger iterations (Ross et al., 2011) beside an exposure-matched continued-SFT control. SFT trains a fresh LoRA adapter (Hu et al., 2022) on 512 replay-verified records for one epoch (16 updates), except the BFWS gate checkpoint, which trained on the full 47,780-record corpus for two epochs. The backbone is Qwen3-VL-8B-Instruct (Bai et al., 2025) (revision 0c351dd0), with InternVL3.5-8B (Wang et al., 2025) as second backbone.

Exact-classical runs the declared algorithm and bounds perfect decisions, random-valid proposes only runtime-valid operations, neither is a learned ability, and when a reference saturates a panel the learned-to-control gain there is unmeasurable rather than zero. Under the enumeration contract, random-valid reads the same runtime-built model input as the policy and, after filtering, draws one operation per decision uniformly at random with a seeded pseudo-random generator from the listed candidates, which are successor-candidate rows under the additive algorithms, non-duplicate evaluated successors with an admissible declared frontier position under BFWS, and unvisited successors under BFS, with a frontier-retire operation under BFWS and BFS when none remains. The sampled operation is submitted once through the same runtime path as a model output, with no retry and no repair. Under the choice-frontier contract the control is uniform menu choice of one offered frontier label per decision under a seeded generator, again with no retry or repair, so the two definitions share a name but not a distribution and each random-valid number is read under the contract of its row.

Splits are by whole task instance with audited corpora (Data and Split Procedures appendix). The modality matrix evaluates 24 tasks from 12 domains under three observations, giving 288 episodes per arm. Budgets count decision calls with invalid operations charged. Learned adapters are capped at twice the exact-reference decision count, the base is priced at one call, and an episode ends at its first invalid emission and counts only after independent replay reproduces it. Contrasts use a paired whole-instance bootstrap (10,000 resamples, seed 1729, 95% percentile) unless a protocol states otherwise (Statistics appendix), and transfer uses exact McNemar tests with Holm correction across 36 comparisons. At these sizes the intervals are descriptive bounds and cannot support broad superiority claims.

The two contracts are compared in Figure 2 (Appendix) and Table 16. The choice-frontier contract has as its pre-registered pass condition the per-pair identity audit of random-valid against exact reference over 18 seed-17 pairs (CHOICE_SENSITIVE on any divergence, ZERO_DECISION_HEADROOM otherwise, no re-runs), also run as a diagnostic on the enumeration contract’s additive cells.

Follow-up protocols (Appendices P–R) were frozen before first outcomes on separate ledgers, adding two image-only native arms that keep only the grounded, unscored menu as text (`visual-nomem-state`, and `visual-seq-state` with $K=8$ predecessor scenes) and menu manipulation by seeded permutation and injection of up to $k=3$ schema-valid inapplicable actions.

The choice-frontier evaluation reuses the frozen 9-task native-arms membership under both additive algorithms, and the same nine tasks (8 domains, 7 compact and 2 expanded variants, optimal costs C^* from 3 to 6) also serve the observation-corruption suite, the native arms, menu manipulation, and the comparator zoo, and the choice-frontier contract and the zoo’s metric set were frozen after earlier outcomes on these tasks (risk and required confirmation in Section 5.11), and the training corpus was derived on CPU by replaying the frozen exact traces through the new contract and verified expansion-for-expansion against them (58 episodes bit-exact).

The CPU-only, exploratory comparator zoo tests Pareto dominance over M1 (solve-versus-budget area over multipliers 1 to 2), M2 (expansion overhead), and M3 (plan cost) against five comparators under a fifteen-test Holm family, with `bfs-order`, `novelty-first` selection, and `worst-first` reading trusted g/h scalars or PDDL atoms the model never sees (so not information-matched), and

with all zoo metrics on the stored choice-frontier episodes post-hoc exploratory and non-significance licensing no equivalence claim.

At the 2026-09-23 snapshot (receipts in Table 4), the BFS pilot closed as history with VALID_STOP, the BFWS gate passed on its development panel, and the modality matrix, DAgger, failure calibration, corruption, transfer, seed replication, and all follow-ups executed. Generalization and second-backbone ran only at reduced v2 scope, because their full-scope v1 branches were withdrawn after their terminal VALID_STOP receipts (closed 2026-09-21) and will not be executed. The BFWS-to-BFS panel overlap is unpinned [TODO: panel overlap, needs the post-#126 retention manifest, issue #131], and the final primary evaluation is pending on undecided prerequisites, so no final comparative efficacy finding exists.

5 RESULTS

This section reports the executed evidence in gate order, with success, validity, invariant compliance, invalid-operation charges, budget, and learned-to-control gain kept as separate measurements and operational detail in Appendices J to S. The random-valid and exact-classical references are oracle-assisted bounds, not learned abilities, and where they saturate they bound what the comparison can measure rather than certify policy quality. Every panel is a development panel, every learned arm rests on one training seed (17) except two replication cells (Section 5.11), and the held-out final evaluation has not been run, so no finding here is a final comparative efficacy claim.

5.1 GATE RECEIPTS AND EXECUTED SCOPE

The expanded study program spent 68.5244 of its 336 GPU-hour cap (2 A100 80GB GPUs) after the corruption and seed-replication addenda, and every executed branch and the four separately ledgered follow-up windows (native arms, stress tests R1-R4, choice frontier, and the CPU-only comparator zoo) reconcile with complete coverage and independent replay, including 1,152/1,152 baseline episodes. Of seven admitted branches, five completed at v1 scope, and two (second backbone, generalization/robustness) stopped with terminal VALID_STOP receipts at their v1 cost gates and completed only at reduced scope under versioned protocols. Their full-scope v1 branches were withdrawn after their terminal VALID_STOP receipts (closed 2026-09-21) and will not be executed. Before the program, the BFS pilot closed with a governed VALID_STOP because random-valid also reached 1.0, the BFWS structural gate passed, the optimal A* arm was retired, and the 45-task held-out manifest was frozen before outcomes and never accessed (Appendix J, Table 7).

5.2 ENUMERATION-CONTRACT IDENTITY AUDIT

The identity audit compares random-valid against exact reference decision by decision on every additive cell of the expanded baseline, because it fixes what the additive rows of the primary matrix measure. Under the enumeration contract the two controls are decision- and expansion-identical on 48/48 additive pairs, because each expansion requires the complete candidate set and heap order ignores submission order, so any policy that emits a listed candidate at every decision follows the exact frontier evolution. Choice quality is unmeasurable by construction on these cells, and their saturating random-valid entries measure operation validity. The audit covers the additive cells only, and the unaudited BFS and BFWS cells fall below saturation. This is an audit finding rather than a performance row, and it is not comparable with the choice-frontier rows of Section 5.9 (mechanism in Appendix R).

5.3 PRIMARY MATRIX: SUCCESS BY ALGORITHM AND CONDITION

The single positive learned-control comparison comes from the frozen 15-task text-state BFWS development panel, where one fixed-seed process-SFT run achieved 0.933 success, versus 0.467 for the oracle-assisted random-valid control, 0.0 for the pretrained base, and 1.0 for exact execution, with a paired whole-instance bootstrap 95% interval lower bound 0.2 on the gain (10,000 resamples). This is a development-stage result from one training seed, and the held-out manifest was not accessed. On the BFS and the two additive best-first development panels random-valid saturates at 100%, so executable behavior there does not identify learned advantage. The expanded base-

line (24 whole problems from 12 domains, four algorithms, three modalities, 576 model episodes independently replayed) gives process SFT 125/288 successes, versus 240/288 for random-valid, 0/288 for the base, and 288/288 for exact execution. Learned success appears only in the additive best-first cells, which the identity audit reads as operation validity. On the BFS and BFWS rows random-valid reaches 15/24 and 17/24 against 24/24 for exact execution, so choice registers there, and process SFT scores 0/24 in every modality. Against random-valid, process SFT wins zero of the 72 problem-modality units in any algorithm column. These are tiny-subgroup descriptive bounds over one training seed, and the development-panel BFWS positive does not reappear on this suite (Appendices K and L, Tables 8 and 9).

5.4 SECONDARY PANELS

Matched modality (process SFT 5/12, 6/12, 6/12 versus 10/12 random-valid per modality) shows no intrinsic modality effect, reduced-scope generalization v2 gives 128/150 learned versus 0/150 base successes under saturating controls, Dagger ties continued SFT at 1/72 each, model-generated successors collapse success from 45/45 to 1/45, all curriculum interaction intervals include zero, and no transfer comparison survives Holm correction (0/36). On InternVL3.5-8B a BFS cell reaches 0/36 and a greedy additive cell 31/36 (base 0/36, random-valid 24/24 per modality), isolating the vision stack rather than the shared Qwen3-8B LLM family (Appendix M).

5.5 FAILURE-MECHANISM CALIBRATION

A frozen descriptive CPU calibration over 2,910 replay-verified episodes classified all 905 failures, with the base failing 447 of 447 episodes on the first decision and the dominant learned kind `other_invariant_violation` (229, 56.3% of learned failures), meaning parsed, source-correct operations rejected by the declared search invariant. Process supervision thus teaches well-formed operations but not invariant-consistent frontier intent over a full episode (Appendix N).

5.6 OBSERVATION-CORRUPTION CHANNEL TESTS

On frozen checkpoints and $k=9$ tasks, copyability-preserving text shuffling lowers success by $-0.944 [-1.000, -0.833]$ (text-state) and $-0.722 [-0.889, -0.500]$ (multimodal), visual corruption costs at most $-0.111 [-0.278, +0.000]$, and text masking makes every episode all-invalid, which is output-contract destruction, so the language-channel claim survives only in its shuffled form. This is a channel finding about the trained controller under the declared adapters, not evidence about intrinsic modality value (Appendix O, Table 13).

5.7 MENU MANIPULATION

On the frozen native adapters (stress test R1, 72 replayed episodes), menu permutation leaves 36/36 successes and 0/512 invalid operations, which is uninformative about order under the submission-order invariance of Section 5.2. Injecting $k=3$ schema-valid but inapplicable actions collapses success to 0/36, with 36/60 invalid operations and a distractor pick-rate of 0.6 against a uniform-picking expectation of 0.592 to 0.625, and every episode ends on a distractor pick, 20/36 on the first decision. The receipt therefore shows that the policy does not reject schema-valid inapplicable menu entries (pick-rate at chance), and these out-of-distribution distractors never appeared in training menus. Content-reading and position habits both predict a pick-rate near 0.6, so this receipt does not separate them, and the fixed-versus-random-position test that would have not been run. Appendix P gives the receipts (Table 14, native arms in Section 5.8).

5.8 IMAGE-ONLY NATIVE ARMS

The native arms see state only as rendered images plus the grounded, unscored candidate menu, and both solve 18/18 clean episodes with zero invalid operations while random-valid is at ceiling, so no structural-advantage claim is made. Their pre-registered corruption payoff is a null (0.000 [0.000, 0.000] for blanking and degrading every image), which the identity audit makes equally consistent with a policy that uses no state information (Appendix Q).

Table 1: First measurement under the choice-frontier contract: frozen 9-task panel, 18 cells (9 tasks $\times$ 2 additive algorithms as repeated measures), 144 episodes independently replayed. Exact reference solves by construction of the budget and is the imitation target. Not comparable with enumeration-contract rows.

Condition	Goal rate	Failures	Invalid ops
Exact reference	18/18	none	0
Learned adapter	7/18	11 budget exhausted	0
Pretrained base	0/18	18 invalid at first call	18
Random-valid (5 frozen seeds)	32/90	58 budget exhausted	0

Table 2: Exploratory pre-registered metric set at the frozen shared $2\times$ cap (9 panel tasks $\times$ 2 additive settings). M1 is the task-clustered solve-versus-budget AUC over $m \in \{1, 1.25, 1.5, 1.75, 2\}$ (9-task bootstrap, seed 133, 10,000 draws). M2 is the solved-only task-weighted geometric expansion overhead ρ and M3 the solved-only task-weighted cost ratio κ , both selection-conditioned. The $3\times$ and $4\times$ columns are additional CPU controls, not M1. Selectors read privileged information that the model does not see.

Arm	M1 AUC [95% CI]	$2\times$ solved	$3\times$	$4\times$	M2 ρ	M3 κ
Learned adapter	.1042 [.0278, .2014]	7/18	n/a	n/a	1.432	1.025
Pretrained base	0 [0, 0]	0/18	n/a	n/a	undefined	undefined
Random-valid	.1389 [.0250, .2722]	32/90	42/90	50/90	1.407	1.000
Exact reference	.8472 [.7917, .8750]	18/18	n/a	n/a	1.000	1.000
bfs-order	.0694 [0, .1528]	6/18	8/18	10/18	1.613	1.000
Novelty-first selection	.1250 [0, .2778]	6/18	8/18	12/18	1.452	1.000
worst-first	.0556 [0, .1389]	4/18	6/18	8/18	1.549	1.050

5.9 CHOICE-FRONTIER REDESIGN

The choice-frontier contract, frozen before launch, lets the policy select which runtime-owned frontier state to expand next, under a budget of $2\times$ the additive exact-reference expansion count with goal selection uncharged. Its pre-registered identity gate passes on 9/9 tasks, a sanity check that holds by construction whenever a menu offers two or more states, and Table 1 gives the first measurement from two seed-17 adapters (Appendix R).

Paired bootstrap contrasts (seed 61813, 10,000 resamples) give random-valid minus exact reference $-0.644 [-0.856, -0.422]$, material, so the controls separate under this contract. The learned adapter minus pretrained base is $+0.389 [+0.167, +0.611]$, material, with zero invalid operations, a contract-fluency gain. The learned adapter minus random-valid is $+0.033 [-0.033, +0.133]$, not material. The 18 cells are 9 tasks with two algorithms as repeated measures, and that last contrast rests entirely on one task (storage), the only one where random-valid’s seed frequency lies strictly between 0 and 1. That falls below the frozen rule that strata under eight cells are descriptive only, so the contrast is descriptive. The evidence supports no learning-to-plan claim for the adapter, and no equivalence claim with random-valid, since no equivalence margin was pre-registered.

5.10 MEASUREMENT CALIBRATION AGAINST A COMPARATOR ZOO

A CPU-only follow-up calibrates this measurement against three deterministic selectors at the same $2\times$ cap (bfs-order, novelty-first selection, and the negative control worst-first), under an exploratory metric set pre-registered before any comparator episode existed, with the stored choice-frontier episodes entering post-hoc. The selectors act on the additive frontier, not native BFS or BFWS, and read privileged scalars or PDDL atoms while the model sees only scenes, so no information-matched learned-versus-rule claim follows (Appendix R).

On M1 the learned adapter falls between bfs-order and random-valid and below novelty-first selection (Table 2, Figure 1), and its M2 and M3 values are selection-conditioned. Across the fifteen-test Holm family no comparison supports positive learned Pareto dominance, and three co-solved tests lack enough support to be defined. Non-significance here is neither equivalence nor a generalized negative capability claim, and the one-task M5 subset is descriptive only (Appendix R).

5.11 DEVELOPMENT-STAGE BOUNDARY

All results above are development-stage, and the held-out final primary evaluation on the frozen 45-task manifest has not been executed, so no final comparative efficacy finding exists. Every learned cell uses seed 17 except two prospective replication cells, aggregated descriptively ($n=3$,

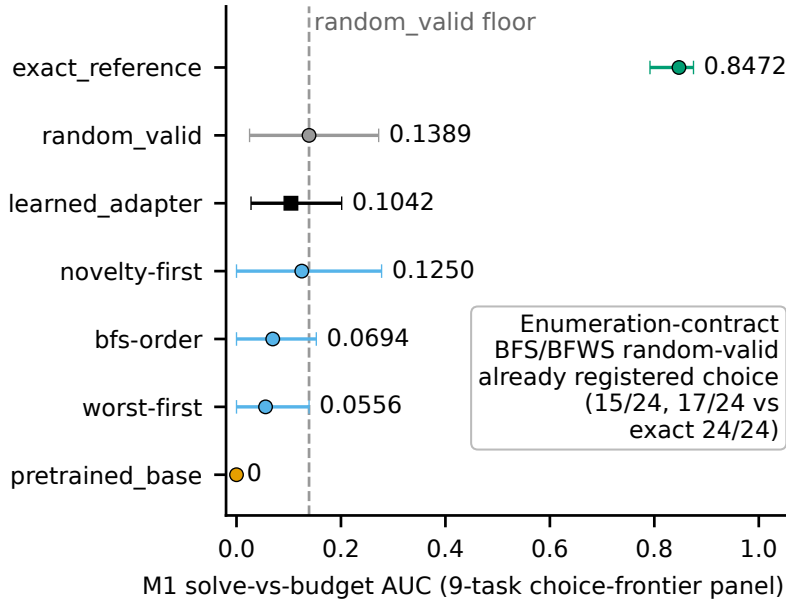


Figure 1: Solve-versus-budget AUC (M1) on the frozen 9-task choice-frontier panel at multipliers 1 to 2. Exact reference .8472, random-valid floor .1389, learned adapter .1042 [0.0278, .2014], novelty-first .1250, bfs-order .0694, worst-first .0556, pretrained base 0. Intervals overlap and the Holm family licenses no ordering. Random-valid is the floor, not a selector.

Appendix S). The full-scope v1 generalization and second-backbone branches were withdrawn after their terminal VALID_STOP receipts (closed 2026-09-21) and will not be executed, and release URLs are pending [TODO: release URLs, issue #122].

One panel of nine low-cost tasks (C^* from 3 to 6) serves the corruption suite, the native arms, menu manipulation (stress test R1), the choice-frontier redesign, and the comparator zoo. The choice-frontier contract was frozen after the stress-test outcomes and the zoo’s metric set after the choice-frontier outcomes on these tasks, so each design step could adapt to earlier results. No held-out task has been evaluated under the choice-frontier contract, and a fresh frozen panel outside these nine tasks, including non-trivial tasks, is required before any instrument-level claim.

6 DISCUSSION AND LIMITATIONS

The results form a measurement finding, the audit that exposed it, and an audit-driven fix. On the additive best-first cells of the enumeration contract, choice quality is unmeasurable by construction, because an expansion must submit its complete candidate set and heap order ignores submission order, while the BFS and width-based best-first cells register choice and the learned policy never succeeds there (Section 5.2). Menu permutation is therefore uninformative about order, and distractor injection shows only that the policy does not reject schema-valid inapplicable menu entries (pick-rate at chance). The choice-frontier redesign separates `random_valid` from `exact_reference`, and its first measurement shows contract fluency rather than choice quality. Its contrast against `random_valid` rests on one discriminative task, so policy-level resolution currently rests on that task, and without a pre-registered margin it licenses neither superiority nor equivalence. The evidence supports no learning-to-plan claim for the redesigned arm.

Every positive result depends on the Trusted Search Runtime, which validates and applies emissions but never chooses operands or repairs them, so decisions are attributable to the model and the evidence does not establish unassisted internal search or general planning autonomy. Within that boundary the output contract and candidate enumeration are learnable, a second backbone replicates this contract fluency on the additive cells, and the one positive learned-control contrast (the BFWS development panel) remains a single-seed development-stage finding. Process supervision yields well-formed operations but not invariant-consistent frontier ordering, and the trained controller depends on the textual serialization only in its copyability-preserving shuffled form. As a post-hoc

interpretation the candidate menu, the input every arm retains, carries the decision-relevant signal on this panel. The DAGger, transfer, and curriculum studies returned nulls, the evidence does not support an intrinsic modality effect, and the comparator zoo finds no positive learned Pareto dominance, which is not a general negative capability claim.

Threats to validity.

- *Development stage.* Held-out manifest unaccessed, single seed except two cells ($n=3$).
- *Saturation.* Saturating controls bound measurability, not policy quality.
- *Exposure.* Longer gate-era training confounds the missing BFWS positive.
- *Imitation target.* `exact_reference` solves by construction of the budget.
- *Comparators.* Zoo selectors read privileged inputs, so ranking is not information-matched.
- *Menu and nulls.* No arm removes the menu, and ceiling nulls do not show image irrelevance.
- *Panel reuse.* Adaptive-overfitting risk on one reused panel requires a fresh frozen panel.

The full-scope v1 generalization and second-backbone branches were withdrawn after their terminal VALID_STOP receipts (closed 2026-09-21) and will not be executed, and release URLs are pending [TODO: release URLs, issue #122].

7 CONCLUSION

Grounded-menu search evaluations can leave choice quality unmeasurable by construction, and cheap identity and menu audits expose this. A redesigned choice-frontier contract separates the controls, and its first measurement shows that the adapter learns to emit valid choices without demonstrating choice quality beyond the named references. These are development-stage findings, not a general efficacy claim, and they establish neither an intrinsic modality effect nor planning autonomy. The next gated step is the pre-registered held-out evaluation on the frozen 45-task manifest, together with a fresh choice-frontier panel (Section 4).

REPRODUCIBILITY STATEMENT

Every stage of the study runs under a frozen outcome-blind protocol. Development panels and held-out manifests are frozen before any model outcome. Pre-registered gates close each stage with a recorded receipt (PASS, VALID_STOP, INVALID, or ANCESTOR_STOP), and no gate was revised after observing results. Training seeds are frozen pre-launch: seed 17 for every cell after the historical pilot, with seeds 29 and 71 added for the two prospective replication cells. The follow-up windows likewise train one seed-17 run per cell unless a protocol states otherwise, and stress tests R1, R2, and R4 run on frozen checkpoints without retraining. No seed was dropped after outcomes. The replay-determinative interface makes every episode independently verifiable from its raw emissions alone. The policy emits Typed Search Operations with explicit operands, and the Trusted Search Runtime validates and applies each operation without filling omitted choices or repairing outputs. Every episode record (model and control) across the seven branches, the observation-corruption suite (360 new + 144 reused bindings), and the seed-replication amendment (534 episodes) was independently replayed with zero missingness. The 1,152 expanded-baseline bindings include 576 model episodes and 576 CPU control/reference episodes.

Every follow-up branch was replayed with the same discipline and zero missingness. The native arms replayed 216/216 new bindings plus 72/72 reused comparators. Stress tests R1, R2, and R4 replayed 170/170 episodes, and the second-backbone v3 cell (stress test R3) replayed 72/72 model episodes plus 72/72 comparators. The choice-frontier evaluation replayed 144/144 episodes, and its 58 preparation episodes were re-derived bit-exact. The comparator zoo replayed 432/432 bindings, 342 of them new CPU episodes.

Two process corpora supply the training data. The BFS text-state corpus contains 90 replay-verified traces and 25,109 process rows. The BFWS corpus covers 105 tasks and 69,019 replay-verified teacher decisions, projecting to a 47,780-train / 21,239-dev split. Both passed no-overlap, no-conflict, live-parity, and token-budget audits before use. Per-cell training configurations, evaluation manifests, LoRA adapter weights with checkpoint digests, and the synthesis claims inventory are recorded for each executed stage.

The program consumed 68.5244 of a capped 336 GPU-hours on 2×NVIDIA A100 80 GB, with a per-branch ledger that retains failed and cutoff attempts at their recorded cost. The follow-up windows ran on separate ledgers with their own caps, and their spend is not added to the program total. The native-arms window spent 3.2086 of 12 GPU-hours, the R1-R4 follow-up window 7.7998 of 36, and the choice-frontier window 1.069 of 12. The choice-frontier corpus derivation and controls ran on CPU, and the comparator zoo ran on CPU only. Paired contrasts use whole-instance bootstrap with 10,000 resamples at seed 1729 and 95% percentile intervals, except where a follow-up protocol fixes its own seed (Statistics appendix). Transfer comparisons use exact McNemar tests with Holm correction across the 36 adapter–base pairs. Strata with fewer than 8 instances are reported as descriptive only. All seeds, panel memberships, corruption parameters, and frozen protocols are stated in the evaluation manifests.

Public release URLs and artifact DOIs have not yet been assigned [TODO: release URLs, issue #122]. An earlier release of the study predates the follow-up evidence, and no successor release is published.

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A OPERATION SCHEMAS

Each declared algorithm defines a vocabulary of operation types τ . A policy emission $u_t = (\tau_t, \theta_t)$ pairs a type from that vocabulary with an operand vector θ_t carrying every exploration-determining argument. The runtime rejects any emission whose type falls outside the declared vocabulary or whose operand vector omits a required field. Every operand that determines exploration must originate from the policy (Section 3).

BFS. The primary operation is node expansion. Its operands name the source state (the current frontier head), the grounded successors selected for insertion, the action producing each successor, and the enqueue position for each. The runtime checks that the named source matches the FIFO frontier head, that each successor results from an applicable grounded action in the source state, that the successor is unvisited, and that each insertion targets the FIFO tail. A goal-test operation names the state to test. The BFS pilot corpus exposes exact grounded successor candidates and full visited membership at each step.

BFWS (complete unpruned). Expansion operands carry the novelty value and goal-count priority of the source state alongside the fields above. Insertion operands carry each candidate’s duplicate membership, novelty, and priority. The runtime checks that the selected source has the declared best priority, that duplicate membership is correctly reported, and that insertions follow the declared ordering. The BFWS corpus exposes candidate duplicate, novelty, priority, insertion, and enqueue facts at every training observation.

Additive best-first. Operands carry the source state’s accumulated path cost g , heuristic estimate h_{add} , and composite value $f = g + w \cdot h_{\text{add}}$ with $w = 3$ for the weighted variant, or $f = h_{\text{add}}$ for the greedy variant. Insertion operands carry per-successor g , h_{add} , and f . The runtime checks that the expanded state has the minimum f in the frontier under the declared tie-breaking rule and that all arithmetic is consistent.

Validity preconditions. An emission passes through a fixed sequence of checks. Failure at any step rejects the emission with the corresponding kind and terminates the episode. The checks, in order:

1. **Parse.** The raw output must parse into a well-formed Typed Search Operation with a recognized type and complete operand vector. Failure: `malformed_output`.
2. **Source identity.** The named source state must match the state the algorithm expects. Failure: `source_state_mismatch`.

Table 3: Rejection taxonomy from the failure-mechanism calibration. Counts are failures per arm class across 2,910 replay-verified episodes (905 total failures, 0 unclassified). Share is within the arm class.

Arm class	Rejection kind	Failures	Share
Base (447 ep., 447 fail.)	malformed.output	379	84.8%
	invalid.frontier.operation	67	15.0%
	other.invariant.violation	1	0.2%
Learned (906 ep., 407 fail.)	other.invariant.violation	229	56.3%
	malformed.output	72	17.7%
	inapplicable.grounded.action	56	13.8%
	effect	27	6.6%
	schema	16	3.9%
	invalid.frontier.operation	5	1.2%
	source.state.mismatch	1	0.2%
	static.context	1	0.2%
Oracle control (51 fail.)	expansion.budget.exhausted	51	100.0%

3. **Action applicability.** Every grounded action in the operands must be applicable in the source state. Failure: `inapplicable.grounded.action`.
4. **Successor schema.** Any named successor state must satisfy the domain schema. Failure: `schema`.
5. **Successor effects.** The named successor must match the declared effects of the applied action. Failure: `effect`.
6. **Static context.** Referenced predicates, objects, and constants must be consistent with the task instance. Failure: `static.context`.
7. **Frontier discipline.** The frontier manipulation must respect the declared ordering (FIFO for BFS, priority for BFWS, f -order for additive best-first). Failure: `invalid.frontier.operation`.
8. **Algorithm invariant.** Any remaining algorithm-specific constraint not covered above. Failure: `other.invariant.violation`.

The two broad categories are *malformed-output* rejections, where the emission never reaches the operation checker, and *post-parse rejections*, where the emission parses but fails a runtime check. The oracle-assisted random-valid control, which emits only runtime-accepted operations, has a single failure kind: `expansion.budget.exhausted`, recorded when the frozen episode budget runs out before a goal is reached.

Rejection counts. Table 3 gives the counts from the failure-mechanism calibration, which classified all 905 failures across 2,910 replay-verified episodes with zero unclassified. The `effect`, `schema`, and `static.context` kinds arise in successor-prediction episodes, where the model generates a dynamic-state successor.

B ALGORITHM INVARIANT DEFINITIONS

The Trusted Search Runtime checks the following invariants on every operation. Stepwise validity and invariant compliance do not establish termination, completeness, optimality, or episode success (Section 3).

BFS. The frontier is a FIFO queue. Every expansion must select the queue head as its source. Every insertion must target the queue tail. A state already in the visited set must not be re-inserted. This invariant is sufficient for completeness in finite state spaces with unit-cost transitions.

BFWS (complete unpruned). The frontier is a priority queue ordered first by novelty rank (lower is better), then by goal count among states with equal novelty. No generated successor is pruned. A successor whose state already appears in the frontier or visited set is not re-inserted.

Additive best-first, weighted ($w = 3$). The frontier is a priority queue ordered by $f(s) = g(s) + 3 \cdot h_{\text{add}}(s)$, where $g(s)$ is the accumulated path cost and $h_{\text{add}}(s)$ is the additive heuristic. The

expanded state must have the minimum f in the frontier. When a successor is generated with a lower g than an existing frontier entry for the same state, the entry is reopened with the improved cost.

Additive best-first, greedy. The frontier is ordered by $h_{\text{add}}(s)$ alone, with no path-cost component. The expanded state must have the minimum h_{add} . Reopen and duplicate rules follow the same convention as the weighted variant.

C TRUSTED SEARCH RUNTIME

The runtime provides three guarantees that hold across all declared algorithms.

1. **No-repair property.** The runtime validates, applies, persists to Search Memory, or rejects each emission. It never chooses omitted operands, reorders candidates, silently repairs an invalid emission, or substitutes a default search decision.
2. **Reject-and-terminate.** On the first rejected emission the episode terminates with the rejection kind recorded. There is no retry path.
3. **Replay determinism.** Replaying the raw emissions $u_{1:T}$ against the same task instance and the same initial Search Memory reproduces the explored trace. Any step whose outcome depends on runtime repair would violate this property.

The procedures below specify the per-step check and the episode loop. No algorithm package is loaded. The listing uses $\text{\LaTeX}$ tabbing.

```

procedure STEP( $u_t, M_{t-1}, A, x$ )
  // Phase 1: Parse the raw emission
  if  $u_t$  does not parse as  $(\tau, \theta)$  with  $\tau \in \text{vocab}(A)$  then
    return REJECT(malformed_output)
  // Phase 2: Validate source identity
  if  $\theta.\text{source} \neq \text{expected}(A, M_{t-1})$  then
    return REJECT(source_state_mismatch)
  // Phase 3: Validate operands against the task and state
  for each grounded action  $a$  named in  $\theta$  do
    if  $a$  is not applicable in  $\theta.\text{source}$  under  $x.D$  then
      return REJECT(inapplicable_grounded_action)
  for each successor state  $s'$  named in  $\theta$  do
    if  $s'$  violates the domain schema then
      return REJECT(schema)
    if  $s' \neq \text{effects}(a, \theta.\text{source})$  then
      return REJECT(effect)
  if any static-context reference in  $\theta$  is inconsistent with  $x$  then
    return REJECT(static_context)
  // Phase 4: Check Algorithm Invariant
  if frontier discipline of  $A$  is violated by  $(\tau, \theta, M_{t-1})$  then
    return REJECT(invalid_frontier_operation)
  if any other invariant of  $A$  is violated then
    return REJECT(other_invariant_violation)
  // Phase 5: Apply and persist
   $M_t \leftarrow \text{apply}(\tau, \theta, M_{t-1})$ 
  return ACCEPT( $M_t$ )

procedure EPISODE( $x, A, \phi_m, B$ )
   $M_0 \leftarrow \text{InitSearchMemory}(A, x)$ 
  for  $t = 1, 2, \dots, B$  do
     $o_t \leftarrow \phi_m(c_t(x, M_{t-1}))$  // build observation
     $u_t \leftarrow \pi(o_t)$  // policy emission
     $r \leftarrow \text{STEP}(u_t, M_{t-1}, A, x)$ 
    if  $r$  is REJECT(kind) then
      return ( $t, \text{FAIL}, \text{kind}$ )
   $M_t \leftarrow r.M$ 

```

```

810         if GoalTest( $M_t, x.G$ ) = TRUE then
811             return ( $t$ , SUCCESS)
812         return ( $B$ , FAIL, expansion_budget_exhausted)
813

```

Each call to STEP consumes one unit of the frozen episode budget B , whether the emission is accepted or rejected. Invalid operations are therefore charged to the budget (Section 4).

D DATA AND SPLIT PROCEDURES

All training and evaluation splits are by whole task instance. No state, decision, or target from an evaluation task appears in any training row. Every released corpus passed no-overlap, no-conflict, live/training-parity, and token-budget audits before use.

BFS pilot corpus. The v6 text-state process corpus holds 25,109 process rows split into 12,994 train and 12,115 dev rows, drawn from 90 replay-verified trusted traces covering 90 distinct tasks. The release assigns splits by semantic task identity, with zero cross-split task overlap. The v8 evaluation panel selects 15 tasks from this corpus’s development set, not its training set. No v8 evaluation-panel task contributes training rows, consistent with the whole-instance rule above. Corpus observations expose the canonical task context, exact grounded successor candidates, and full visited membership. Release audits recorded zero semantic, input, and input-target overlap, zero conflicting identical-input labels, zero teacher-operation rejections, and zero live/corpus mismatches, all within an 8,192-token input cap and a 384-token output allowance.

BFWS corpus. The corpus covers the frozen 105-task development panel with 69,019 replay-verified teacher decisions. Its release contains 69,019 process records, 67,215 operational records, and a process projection of 47,780 train and 21,239 dev rows. Each step’s observation exposes candidate duplicate membership, novelty values, goal-count priorities, insertion positions, and enqueue facts. Audit counters recorded zero overlap, mismatch, rejection, or held-out-access violations.

Evaluation manifests. The BFS pilot uses a frozen 15-task outcome-blind development panel. The BFWS structural gate froze a 105-task development panel and a separate 45-task held-out manifest. The held-out manifest has never been accessed, and no model prediction, training decision, or analysis references it. The expanded baseline evaluates 24 tasks from 12 domains under 4 algorithms, 3 modalities, and 4 arms (1,152 logical bindings, 576 model episodes, all independently replayed). The curriculum stage evaluates 27 unseen tasks. The generalization-v2 protocol admits 25 derived tasks from 93 eligible semantics-preserving variants (uniform $k=5$ cheapest-prefix per family, membership frozen before execution).

E RECEIPT TABLES

Table 4 gives the per-branch gate receipts from the synthesis ledger. Five branches executed to completion at v1 scope. Two stopped at their frozen v1 cost-admission gates with terminal VALID.STOP receipts and later completed at reduced scope under new v2 protocols. Every branch reconciles with zero missing evidence. The Tickets column indexes each row to its internal ticket number.

At the synthesis snapshot, the program recorded 135 attempts (96 succeeded, 36 failed, 3 cutoff). Failed and cutoff attempts retain their hours in the ledger. Every attempt ended before the absolute scheduler cutoff of 2026-09-21T11:55:19Z.

The observation-corruption suite added 2.22 GPU-h to the generalization branch. The training-seed replication addendum (closed 2026-09-21) added 12.79 GPU-h and 534 independently replayed episodes covering two cells. Together these addenda brought the program total to 68.52/336 GPU-h. The Statistics appendix reports the seed-variance results.

Follow-up windows. Each follow-up window keeps its own ledger, and none is added to the program’s 68.52/336 GPU-h. The native-arms window (v1 probes, then v2 training and evaluation) spent 3.2086 of 12 GPU-h. Its 216 new bindings (144 model episodes and 72 CPU random-valid

Table 4: Per-branch gate receipts for the expanded study program (through the corruption and seed-replication addenda) and the follow-up windows. Program spending is 68.52 of the 336 GPU-h cap. Entries and the total are rounded independently from the ledger. One prospective budget transfer (12.11 GPU-h from recovery reserve to second backbone, ratified 2026-09-19) is reflected in the branch caps. The seed-replication row draws on existing baseline and DAGger caps, not an additional allocation. Rows below the total are follow-up windows, each on its own ledger with its own cap, listed at ledger precision and never added to the total. The R1-R4 window spend covers both stress-test rows.

Branch	Tickets	Terminal state	Coverage	GPU-h / cap
Expanded baseline	#116–#118	complete	1,152/1,152	6.53 / 56
DAGger	#78–#84	complete	405/405	10.91 / 48
Successor prediction	#85–#89, #99	complete	90/90	10.85 / 64
Curriculum by modality	#119	complete	567/567	11.78 / 48
Transfer	#104–#107	complete	39/39	3.60 / 24
Second backbone	#101–#103, #123	v2 complete	144/144	6.03 / 52.11
Generalization/robustness	#96, #98, #124, #126	v2 complete	1,200 + 360	6.04 / 32
Recovery reserve	–	reduced	–	0.00 / 11.89
Training-seed replication	#127	complete	534/534	12.79 / –
Total				68.52 / 336
<i>Follow-up windows (separate ledgers, not summed with the total)</i>				
Native arms	#128, #129	v2 complete	216/216 + 72/72	3.2086 / 12
Stress tests R1-R4	#130	complete	170/170	7.7998 / 36
Second backbone v3 (R3)	#130	complete	72/72 + 72/72	same window
Choice frontier	#132	CHOICE_SENSITIVE	144/144	1.069 / 12
Comparator zoo	#133	complete	432/432	CPU only

episodes) and 72 reused comparators were all independently replayed. The R1-R4 follow-up window spent 7.7998 of 36 GPU-h across two ledgers. It was resized from 20 GPU-h before any outcome, because its frozen worst-case estimate of 27.04 GPU-h including spend exceeded the initial cap, and realized cost landed at 29% of that estimate. The choice-frontier window spent 1.069 of 12 GPU-h (training 0.81, smoke 0.052, model evaluation 0.208), with corpus derivation and controls on CPU. Its 144 panel episodes were replayed and its 58 preparation episodes re-derived bit-exact. The comparator zoo launched no GPU job. It replayed all 432 zoo bindings, of which 342 are new CPU episodes and 90 are reused $2\times$ random-valid controls. Failed attempts retain their hours in every window.

Artifact index. Table 5 maps each pinned follow-up evidence path to the class of claim it supports. Paths are relative to the evidence repository.

F STATISTICS

Default contrast procedure. Contrasts are paired by whole task instance. Paired differences use a bootstrap with 10,000 resamples at seed 1729 and 95% percentile intervals. Under one training seed and panels of 24 or fewer instances, these intervals are descriptive bounds. They do not support broad superiority claims.

The BFWS gate used confidence level 0.95 with 10,000 paired whole-instance resamples (bootstrap seed 1729). Its reported lower bound is the 0.025 quantile, i.e., the lower endpoint of a two-sided 95% percentile interval.

Tiny-strata rule. Strata with fewer than 8 episodes are reported as descriptive only. No intervals are computed over such strata. They are listed explicitly and never silently pooled or dropped.

Transfer tests. Transfer comparisons use McNemar exact tests against the per-example base predictions, with Holm correction across all 36 adapter-by-benchmark comparisons. The minimum adjusted p in the executed transfer branch is 0.8156. None survives correction.

Follow-up bootstrap conventions. The follow-up windows fix their own bootstrap seeds, separate from the main program’s seed 1729 default. The native-arms contrasts and the choice-frontier contrasts use a paired whole-problem bootstrap at seed 61813 with 10,000 resamples and 95% percentile

Table 5: Artifact index for the follow-up evidence. Each row names a pinned path and the claim class that rests on it.

Pinned path	Claim class
outputs/native-arms/v1/evaluation/ (evaluation.json, analysis.json)	Native-arms clean and corruption success, paired contrasts
outputs/native-arms/v1/identity-audit. json	Identity finding on the enumeration contract
outputs/native-arms/v1/ stress-evaluation.json	Menu manipulation, degraded strains, R2 cross cell, stress replay
outputs/native-arms/v1/r2-decomposition. json	Text-corruption termination modes
outputs/native-arms/v1/budget.json and the R1-R4 window ledgers (v1, v2)	Native-arms and R1-R4 window ledgers
outputs/expanded-study/v1/ second-backbone-v2/evaluation/ outputs/expanded-study/v1/ second-backbone-v3/evaluation/ outputs/choice-frontier/v1/evaluation/ (evaluation, identity-audit, analysis, final-audit)	Second-backbone BFS cell Second-backbone additive replication and its contrasts
outputs/choice-frontier/v1/preparation/ visual-choice-frontier/ (report, audit)	Choice-frontier gate, panel rows, contrasts, re- play
outputs/choice-frontier/v1/budget.json	Choice-frontier corpus derivation
outputs/choice-frontier/o4/zoo/manifest. json	Choice-frontier window ledger
outputs/choice-frontier/ o4/metrics/ (analysis, stored-posthoc-analysis, optimal-costs, native-four-algorithm-reference-costs)	Zoo coverage and replay
docs/experiments/native-arms/ issue-129-closeout.md, issue-130-closeout.md	Exploratory M1–M5 metrics, Holm family, optimal costs, native reference costs
docs/experiments/choice-frontier/ issue-132-closeout.md	Native-arms and stress-test R1-R4 records
docs/experiments/choice-frontier/ metric-adjudication-2026-09-23.md	Choice-frontier outcome and its binding erra- tum
Comparator-zoo ticket (#133) closing comment	Post-close metric adjudication
	Zoo numbers table and claim boundary

intervals. The second-backbone v3 contrasts (stress test R3) use seed 1729 with 10,000 resamples. The comparator-zoo metrics use a task-cluster bootstrap that draws nine tasks with replacement at seed 133 for 10,000 draws. Both additive cells and all random seeds stay inside their task cluster.

Materiality and non-equivalence. A follow-up contrast is material when its 95% interval excludes zero. No follow-up protocol pre-registered an equivalence margin, so a non-material contrast supports neither superiority nor equivalence. Wherever the text reports a non-material contrast or a comparison that fails Holm correction, an adjacent sentence states that non-significance is not equivalence.

Tiny strata in the follow-ups. The fewer-than-8 rule also governs discriminative subsets. The choice-frontier learned-minus-random-valid contrast is computed over 18 cells, but it rests entirely on storage, the only panel task where random-valid’s seed frequency lies strictly between 0 and 1 (0.2). A one-task discriminative subset falls under the rule, so that contrast is descriptive. The zoo’s M5 subset, defined from controls only, holds the same single task and is reported descriptively.

Zoo dominance tests. Any positive learned-versus-comparator dominance claim in the comparator zoo uses paired task-cluster bootstrap one-sided p values with Holm step-down correction over five comparators and three metrics, fifteen tests in all. Undefined or uninformative co-solved contrasts prohibit a positive dominance claim and never count as wins. All zoo metrics are exploratory, and those computed on the stored choice-frontier episodes are post-hoc.

Table 6: Seed-variance replication for Cell A (headline process-SFT cell). Contrasts are paired whole-instance bootstrap (24 tasks, 10,000 resamples, seed 1729). The primary contrast (SFT – base) is positive at every seed with no sign flip.

Seed	Successes / 24	SFT – base	SFT – random-valid
17	21/24	+0.875	−0.125
29	21/24	+0.875	−0.125
71	22/24	+0.917	−0.083

Seed-variance addendum. A prospective training-seed replication measured variance for exactly two cells. Seeds 17, 29, and 71 were frozen before any seed-29 or seed-71 artifact existed. All 534 new episodes were independently replayed. Cross-seed aggregation is descriptive ($n = 3$ seeds).

Cell A is the headline process-SFT cell (`best_first_add_greedy` × `multimodal-state`) evaluated on the 24-problem panel. Each seed trained a fresh LoRA adapter with new initialization and dropout randomness on the identical frozen 512-record membership. Table 6 gives the per-seed results.

Cell set B covers DAGger iteration-1 versus the exposure-matched continued-SFT control under BFS across all three modalities. The null replicates: 0/72 unseen successes for every arm and every seed. The paired contrast (DAGger minus continued SFT) is +0.000 [0.000, 0.000] at each seed.

This addendum supersedes the blanket single-training-seed boundary for these two cells only. Every other cell in the program remains single-seed (seed 17) evidence.

G MODALITY ADAPTERS AND OBSERVATION CONTRACT

Three fixed Modality Observation adapters are declared. Each maps a decision context c_t (the canonical task context plus the decision-relevant view of Search Memory M_t) to an observation $o_t = \phi_m(c_t)$ presented to the policy. The adapter is fixed for an entire training or evaluation cell. The adapter contract requires decision sufficiency: o_t must expose every fact the exact teacher uses to resolve step t under the declared algorithm.

text-state. The state, goal, and search context are serialized as a compact relational representation. Object identities, predicate names, and all Search Memory contents (frontier, visited set, algorithm-specific state) appear as structured text.

visual-state. The state and goal are rendered as a pair of 128×128 pixel images: a rendered state image and a partial-goal constraint image. Object labels are not embedded in the rendered image. Search Memory contents and the grounded candidate list remain in the textual serialization, even in this adapter. This is the published frozen `visual-state` arm that the native arms below are compared against.

multimodal-state. The observation combines the text-state serialization and the visual-state rendered images. The policy receives the relational text and the image pair jointly.

Native image-only arms. Two further adapters executed under the native-arms v2 contract. `visual-nomem-state` keeps only the static task interface and the grounded, unscored candidate menu as text, and carries state facts in the current rendered scene alone. `visual-seq-state` adds up to $K=8$ rendered history frames, each labelled with its absolute path depth. Neither receives textual Search Memory, accepted deltas, g/h /priority scalars, candidate scores, or membership flags.

Text-scaffold caveat. Search Memory contents and the candidate list are serialized as structured text in all three declared adapters, including `visual-state`. Rendered images carry the planning state and goal visuals, but search bookkeeping (frontier, visited set, novelty counters, candidate list) always travels through the text channel. The observation-corruption suite is consistent with this. Copyability-preserving text shuffling drops success by up to −0.944 (text) and −0.722 (multimodal), while visual corruption costs at most −0.111 with intervals that include zero.

The native arms narrow this caveat to the candidate menu. With the textual bookkeeping and scalars removed, both arms solve 18/18 clean tasks with zero invalid operations, and blanking or degrading

every image of the seq arm gives a corrupted-minus-clean contrast of 0.000 [0.000, 0.000]. As a post-hoc interpretation, the grounded unscored menu is the remaining text channel that could carry the signal. Menu manipulation shows that the policies do not reject schema-valid inapplicable menu entries (pick-rate at chance), and on the additive cells submission-order invariance makes any listed candidate a correct emission. The caveat therefore still holds for every adapter that presents the menu, and no executed adapter isolates an image-only decision channel.

Information-loss boundaries. The 128-pixel unlabelled rendered views can lose symbolic identities. The P3 name-compression family in the generalization-v2 suite is information-lossy. Family-wide (24 eligible P3 variants), 24/24 variants are text-lossy, 18/24 are multimodal-lossy, and 0/24 are visual-lossy. Modality matching between training and evaluation ensures identical observation format, not information parity. Static-context and goal fields of the observation contain text in every adapter. The available evidence does not establish semantic parity across adapters or an intrinsic modality effect.

H RETAINED-INFRASTRUCTURE FEASIBILITY NOTES

The terms below describe verified infrastructure components that support corpus generation, evaluation, and evidence production. They constitute feasibility evidence for the runtime and rendering pipeline. They cannot support a Search Process Policy efficacy claim.

Plan pipeline. **Plan Submission** transmits a supplied plan from the project adapter to a planning or rendering backend. **Plan Interpretation** converts the submitted plan into the backend’s ordered action representation. An **Action Sequence** is the canonical identity of a supplied plan as a normalized, ordered sequence of grounded action predicates. Casing and whitespace are not part of its identity.

Rendering pipeline. **Render Production** creates a visualisation output and its rendered image for a planning state. **Render Validation** confirms that the produced visualisation and image satisfy declared structural and semantic checks. **Plan Provenance** links a produced visualisation to the exact supplied plan that the backend interpreted.

Certification and evidence. **Integration Certification** is the finding that Plan Submission, Plan Interpretation, Plan Provenance, Render Production, Render Validation, and required execution-isolation checks all passed. An **Attempt** is one authorized, immutable execution with its own identity and retained outcome, whether successful or not. An **Evidence Bundle** is the self-contained records from an Attempt that allow an independent reproducer to verify its claims offline. The retained infrastructure supplies deterministic compact delta/event evidence with semantic replay while preserving search semantics, serving the replay-determinism requirement of the Trusted Search Runtime.

Historical terms. Planning Certificate, Joint Action-and-Certificate SFT, Adaptive Scaffolding, Support Route, Live Memory, Verified Joint Step, and Route Label are historical CGAS terms. They describe retained infrastructure internals. They cannot support any current Search Process Policy efficacy claim and are not referenced elsewhere in this manuscript.

I CHOICE-FRONTIER CONTRACT AND COMPARATOR SELECTORS

Menu and observation. At each decision the runtime exposes the live frontier as an unscored menu of opaque labels $c_0, \dots, c_K$. Labels are assigned after a seeded Fisher–Yates permutation of the generation-serial member order, with a per-decision seed derived from master seed 51131 and a hash of the member state identifiers, so a label carries no positional information. The observation holds static task-context pages, the unlabelled 128px initial-state scene, one unlabelled 128px scene per frontier state in menu order, and the partial-goal pages. Its text payload carries only the algorithm name, the label list, the representation tag, the schema version, and the frozen legend. A leak check fails closed on any forbidden key marker.

Output and validation. The policy emits canonical JSON with the single key `expand_choice` and a label value. The runtime rejects malformed JSON, a wrong key set, a non-string label, a label not offered by the pending menu, and a label bound to a non-live member. The first rejection ends the episode as `deterministic_invalid_operation`. Other terminations are `goal_reached`, `frontier_exhausted`, `decision_budget_exhausted`, and `expansion_budget_exhausted`.

Expansion and budget. On acceptance the runtime removes the chosen member from the frontier and expands it atomically. Candidate generation, serial assignment, admission, duplicate detection, reopening, and expansion bookkeeping are inherited unchanged from the frozen additive best-first controller, which is not modified. Per-candidate admissions are runtime-internal, so one submitted choice is one decision. Selecting a member that satisfies the goal solves the task without an expansion and without budget charge, so the exact episode takes its reference expansion count plus one decisions. Decision and expansion caps both equal $2 \times$ the reference expansion count.

Corpus derivation. The training corpus was derived on CPU by replaying the frozen exact traces through the new contract and verified expansion-for-expansion against them (58 episodes bit-exact). The frozen matched-modalities membership indexes enumeration-contract decisions. Each such decision maps to the expansion whose candidate set it submitted into, unique (task, expansion) pairs in first-occurrence order form the base set, and a round-robin over the frozen task order tops the set up to 512 training records per algorithm. Diagnostics are the unique expansions of the frozen diagnostic identifiers, with no top-up. The teacher target is the label bound to the heap head. Five fail-closed gates run before training. They check expansion-for-expansion agreement with the stored exact traces, a token budget of measured input plus 384 within 32,768, the text-leak check, image resolution, and identity with the frozen membership. The pinned preparation report records 29 tasks, 1,046 records (512 training and 11 diagnostic per algorithm), 3,544 scenes, and 58 episodes re-derived bit-exact, with outcome PASS. The smoke gate accepted 40/40 model calls.

Training and replay. Each of the two cells trains a LoRA adapter (rank 64, α 128) for 16 updates on its 512 records at seed 17 with the frozen expanded-program recipe. Both training jobs completed all 16 steps and saved final adapters, then failed in post-save summary construction. Their reports were recovered from the primary training-state artifacts with provenance marked, no retraining was needed, and the training audit passes on the recovered reports. At finalize, independent replay re-derives every menu binding, payload, and trusted transition from the stored events, and checks pixel bindings against the retained scenes.

Comparator selectors. Each selector’s choice is mapped back to the menu binding, and every candidate still enters through the additive admission path. Frontier admission, duplicate detection, h_{add} computation, reopening, and serial ordering therefore stay additive for `bfs_order` and novelty-first selection alike. The novelty features of novelty-first selection are positive PDDL state atoms, and its seen-feature sets advance only when a state is expanded, including the initial expansion. Each new episode is re-run against an independently constructed PDDL authority, which checks menus, decisions, admissions, and result dictionaries. Selector episodes replay the model input, menu, actions, admissions, and budget, but they neither generate nor replay image bindings. The optimal costs C^* come from one offline shortest-plan computation per task with replay-verified plans, and the native four-algorithm reference costs are recorded separately from the zoo.

Metric computation. Under M1 an episode solves at multiplier m when its first goal selection occurs within $\lfloor mR_t \rfloor$ decisions with at most $\lfloor mR_t \rfloor$ preceding expansions, where R_t is the additive exact-reference expansion count. Unsolved $2 \times$ runs count as failures at every m , and the same prefix truncation applies to every arm. Task and algorithm weights are equal. M2 also has a censored form, in which failures are right-censored at the budget rather than assigned a fictitious expansion count. M3 counts transitions along the selected goal state’s trusted predecessor chain and is undefined, never imputed, for unsolved episodes. M4 measures agreement with the current additive heap head over decisions with at least two live choices, against a chance rate equal to the decision-weighted mean of $1/K$. M5 restricts to tasks where the mean $2 \times$ random-valid solve indicator lies strictly between 0 and 1.

Table 7: Gate receipts for the expanded study program and its follow-up windows, before effect interpretation. Terminal states and frozen scopes are copied from the branch reconciliation ledger and the follow-up ledgers. The two reduced-scope branches closed on v2 evidence, and their withdrawn v1 scopes are retained as VALID_STOP history. Rows below the middle rule are follow-up windows on separate ledgers. The second column indexes each row to its internal ticket number.

Branch	Tickets	Terminal state	Frozen scope executed	Coverage
Expanded baseline	#116–#118	complete	1,152 bindings, 576 model episodes	1,152/1,152 replayed
Dagger	#78–#84	complete	405 episodes + 81 paired rows	405/405
Successor prediction	#85–#89, #99	complete	90 episodes, 15 modality-panel cells	90/90
Curriculum by modality	#119	complete	243 model + 324 comparator bindings	all present
Transfer	#104–#107	complete	39 cells: 12 v1 + 27 v2	39/39
Second backbone	#101–#103, #123	v1 VALID_STOP, complete	v2 72 model episodes + comparators	72/72
Generalization/robustness	#96, #98, #124	v1 VALID_STOP, complete	v2 1,200 episodes: 300 model + 900 controls	1,200/1,200
Observation corruption	#126	complete	360 new bindings	360/360 replayed
Training-seed replication	#127	complete	534 new episodes	534/534 replayed
Native arms	#128, #129	v1 gate history, v2 complete	216 new bindings + comparators	216/216 + 72/72
Stress tests R1/R2/R4	#130	complete	170 episodes: 72 menu, 80 variant, 18 cross-cell	170/170
Second backbone v3 (stress test R3)	#130	complete	72 model episodes + comparators	72/72 + 72/72
Choice frontier	#132	complete, CHOICE_SENSITIVE	144 panel episodes, preparation episodes re-derived	58 144/144
Comparator zoo	#133	complete, CPU only	432 zoo bindings, 342 new episodes	432/432

J RESULTS DETAIL: GATE RECEIPTS AND QUALIFICATION GATES

The expanded study program ran on 2 A100 80GB GPUs under a 336 GPU-hour cap with an absolute scheduler cutoff, and spent 53.52 of those hours (15.9%) at the synthesis snapshot, before the corruption and seed-replication addenda, and 68.5244 after them. Seven branches were admitted. Five executed to completion at v1 scope, and two (second backbone, generalization/robustness) stopped with terminal VALID_STOP receipts at their frozen v1 cost-admission gates and later completed at reduced scope under new versioned protocols. Table 7 lists the per-branch receipts with their frozen scopes. Every executed branch reconciles with zero missing evidence: 1,152/1,152 baseline episodes independently replayed, 405/405 Dagger episodes, 90/90 successor episodes, 243/243 curriculum model episodes with 324/324 comparator bindings, 39/39 transfer cells, 72/72 second-backbone-v2 model episodes with 72 hash-pinned comparators, and 1,200/1,200 generalization-v2 episodes. The full-scope v1 branches of the two reduced branches were withdrawn after their terminal VALID_STOP receipts (closed 2026-09-21) and will not be executed. Their terminal VALID_STOP evidence is retained as history.

Four follow-up windows executed after the program, each on its own GPU ledger. These windows are reported separately and never summed with the main program ledger. The native-arms window spent 3.2086 of 12 GPU-hours, the R1-R4 follow-up window spent 7.7998 of 36, and the choice-frontier window spent 1.069 of 12. The comparator zoo ran on CPU only. The R1-R4 follow-up window was resized from 20 to 36 GPU-hours before any outcome, because its frozen worst-case estimate of 27.04 GPU-hours including spend exceeded the initial cap, and realized cost landed at 29% of that estimate. Each follow-up closed with complete coverage and independent replay (Table 7).

Three earlier gates frame the active algorithm matrix. The bounded BFS pilot closed with a governed VALID_STOP: on a frozen outcome-blind 15-task BFS development panel, final checkpoints

Table 8: Frozen 15-task text-state BFWS development panel: full-episode budgeted success by condition. The learned-to-best-control gain over random-valid is 0.467 with a paired whole-instance bootstrap 95% interval lower bound 0.2 (10,000 resamples). Single fixed training seed, development stage, held-out manifest not accessed.

Condition	Success (15 tasks)
Pretrained base	0.0
Process SFT	0.933
Random-valid (oracle-assisted control)	0.467
Exact execution	1.0

Table 9: Primary matrix: expanded 24-task baseline by algorithm and modality. Measures are reported separately and never combined. Columns give full-episode budgeted success out of 24 for each condition, process-SFT invalid operations over decisions (inv/dec), process-SFT expansions, and exact-reference decisions as the budget reference. The pretrained base terminates on its first invalid operation in every cell (24/24 inv/dec, 0 expansions). Random-valid (RV) is an oracle-assisted programmatic control and exact execution is the oracle upper bound, and both are references rather than model abilities.

Algorithm	Modality	Base	SFT	RV	Exact	SFT inv/dec	SFT exp.	Exact dec.
BFS	text	0/24	0/24	15/24	24/24	24/66	15	1969
BFS	visual	0/24	0/24	15/24	24/24	24/69	17	1969
BFS	multimodal	0/24	0/24	15/24	24/24	24/63	15	1969
BFWS	text	0/24	0/24	17/24	24/24	24/24	0	974
BFWS	visual	0/24	0/24	17/24	24/24	24/24	0	974
BFWS	multimodal	0/24	0/24	17/24	24/24	24/24	0	974
Weighted A* ($w=3$)	text	0/24	20/24	24/24	24/24	4/609	179	694
Weighted A* ($w=3$)	visual	0/24	18/24	24/24	24/24	6/499	143	694
Weighted A* ($w=3$)	multimodal	0/24	22/24	24/24	24/24	2/666	196	694
Greedy best-first	text	0/24	22/24	24/24	24/24	2/614	179	673
Greedy best-first	visual	0/24	22/24	24/24	24/24	2/643	189	673
Greedy best-first	multimodal	0/24	21/24	24/24	24/24	3/617	179	673

from five process-SFT seeds achieved 1.0 invariant-valid success with a 0.0 invalid-operation rate, but the base achieved 0.0 and random-valid achieved 1.0, so the learned policy showed executable BFS behavior with zero gain over the strongest control and downstream work was gated off. The BFWS structural gate then passed on its frozen development panel (Section 5.3). The companion 45-task held-out manifest was frozen before outcomes and has never been accessed. The optimal A* arm over $h_{\max}$ with landmark counting was retired after its own governed VALID_STOP and is superseded by the two additive best-first variants. Its original heuristic-representation estimand remains unidentified. The historical five-seed BFS pilot predates the current budget rule, which authorizes one training run at seed 17 per model and cell (Section 4).

K RESULTS DETAIL: PRIMARY-MATRIX TABLES

Table 8 gives the BFWS development panel and Table 9 the expanded 24-task baseline by algorithm and modality, both summarized in Section 5.3. Measures are reported separately and never combined.

L RESULTS DETAIL: PRIMARY-MATRIX PAIRED CONTRASTS

Paired whole-problem contrasts supply the learned-to-best-control gain, kept separate from the success counts (Table 10). Against the base, process SFT gains only in the additive cells (+0.833 for weighted A*, +0.903 for greedy) and gains nothing under BFS or BFWS. Against random-valid, process SFT loses in every algorithm column and wins zero of the 72 problem-modality units in any column. Random-valid itself ties exact execution in both additive columns, so those cells cannot measure any advantage beyond valid bookkeeping. These contrasts are tiny-subgroup descriptive bounds over 24 problems and one training seed. They do not support modality or algorithm superiority claims beyond the executed panel. The development-panel BFWS positive does not reappear on this suite. The complete six-measure matrix for the frozen final protocol, including per-cell Algorithm Invariant compliance and charged-budget columns, belongs to the held-out evaluation.

Table 10: Paired whole-problem success contrasts on the expanded baseline (10,000 paired resamples over the 24 problems, seed 1729). Entries give the point contrast with wins/losses/ties over the 72 problem-modality units. Contrasts are descriptive bounds over a tiny subgroup and a single training seed (Efron, 1979).

Contrast	BFS	BFWS	Weighted A* ($w=3$)	Greedy
SFT – base	+0.000 (0/0/72)	+0.000 (0/0/72)	+0.833 (60/0/12)	+0.903 (65/0/7)
SFT – random-valid	−0.625 (0/45/27)	−0.708 (0/51/21)	−0.167 (0/12/60)	−0.097 (0/7/65)
SFT – exact	−1.000 (0/72/0)	−1.000 (0/72/0)	−0.167 (0/12/60)	−0.097 (0/7/65)
Random-valid – exact	−0.375 (0/27/45)	−0.292 (0/21/51)	+0.000 (0/0/72)	+0.000 (0/0/72)

Table 11: DAgger on BFS: development and unseen success with unseen invalid-operation rates, against the exposure-matched continued-SFT control and the two references.

Arm	Dev success /9	Unseen success /72	Unseen invalid rate
DAgger iteration 2	3/9	1/72	0.316
Continued SFT iteration 2	2/9	1/72	0.175
Original process SFT	1/9	0/72	0.364
Random-valid	6/9	45/72	0.000
Exact reference	9/9	72/72	0.000

M RESULTS DETAIL: SECONDARY PANELS

Modality. The per-modality rows of Table 9 show the same qualitative profile in all three adapters: zero learned success under BFS and BFWS, learned success below the saturating controls in the additive cells. The earlier matched-modality study, a 3-problem panel on its own ledger that is never pooled with the expanded baseline, gives process-SFT success of 5/12 in text-state, 6/12 in visual-state, and 6/12 in multimodal-state, versus 10/12 for random-valid in each modality, 0/12 for the base, and 12/12 for exact execution. The evidence does not support an intrinsic modality effect. Two caveats bound any such claim: the unlabelled 128px rendered views can lose symbolic identities, and the static and goal pages still contain text in every modality, so semantic parity across adapters is not established.

Generalization and robustness v2. The reduced v2 protocol admitted a uniform cheapest-prefix sample of 5 variants per family over 5 perturbation families (25 of 93 eligible semantics-preserving variants, membership frozen before execution) and evaluated the learned additive-best-first adapters. Learned adapters succeeded on 128/150 episodes versus 0/150 for the pretrained base, where every base episode is a one-call invalid termination (150 decisions total). By family, learned success was 29/30 on object-renaming, 27/30 on name-compression, 27/30 on render-restyle, 26/30 on scale-up, and 19/30 on shifted-init. By modality the split was text 44/50, multimodal 43/50, visual 41/50. By algorithm it was greedy 67/75 and weighted A* 61/75. All 22 learned failures are single-invalid-operation terminations, 11 of them in shifted-init. Controls saturate (random-valid 750/750, exact 150/150) and remain bounds, not learned ability. The lossy name-compression stratum is reported separately and never pooled: all 5 admitted variants are text-lossy, 3 multimodal-lossy, and 0 visual-lossy, with learned success on that stratum of text 9/10, visual 8/10, multimodal 10/10. Family strata hold 5 problems each, so family-level rates are descriptive only. The full-scope v1 generalization branch was withdrawn after its terminal VALID_STOP receipt (closed 2026-09-21) and will not be executed.

DAgger. Two DAgger iterations (Ross et al., 2011) on BFS were compared against an exposure-matched continued-SFT control on 3 development and 24 unseen modality-task pairs (Table 11). On the unseen panel, DAgger and continued SFT each achieve 1/72 invariant-valid successes (paired over whole problems: one DAgger win, one loss, 70 ties), both far below random-valid at 45/72 and exact execution at 72/72, while the original process-SFT checkpoint achieves 0/72. The result is null. DAgger also carries the higher unseen invalid-operation rate of the two iterated arms (0.316 versus 0.175). Final cumulative unique training corrections were text 201, visual 203, multimodal 181, accumulated over 12 training cells of 512 records and 16 optimizer updates per cell-iteration.

Model-generated successors. Replacing trusted successors with model-generated dynamic-state successors collapses goal success from 45/45 to 1/45 on the evaluated panel, with zero trusted-state substitutions in the trusted arm. The single success is the one multimodal unseen cell. The trained

Table 12: Zero-shot transfer accuracy on frozen benchmark subsets (FOLIO 200, GSM8K 200, HumanEval 164). Adapter rows follow the transfer ledger names: `bfs`, `iw` (width-based best-first, BFS), `astar_w3`, and `astar_greedy`, each in text (t), visual (v), and multimodal (m) training cells. No comparison against the base survives Holm correction (0/36, minimum adjusted p 0.8156).

Adapter	FOLIO	GSM8K	HumanEval
Base	0.600	0.955	0.780
<code>bfs t/v/m</code>	0.600 / 0.605 / 0.605	0.950 / 0.945 / 0.965	0.793 / 0.799 / 0.780
<code>iw t/v/m</code>	0.660 / 0.645 / 0.645	0.960 / 0.965 / 0.960	0.768 / 0.762 / 0.756
<code>astar_w3 t/v/m</code>	0.590 / 0.575 / 0.590	0.950 / 0.965 / 0.945	0.817 / 0.799 / 0.774
<code>astar_greedy t/v/m</code>	0.590 / 0.590 / 0.590	0.950 / 0.955 / 0.950	0.799 / 0.811 / 0.793

successor model is exactly correct on 65/109 retained raw predictions, with 93 schema-valid and 66 effect-valid, and downstream failure kinds are dominated by exact state-identity and effect failures (27 effect, 16 schema, 1 static-context) rather than schema failures. Collection-time verification of the untrained exact-successor accepted 119/1,536 predictions (text 43, visual 52, multimodal 24). Coverage is partial by design: the frozen outcome-blind admission accepted a 12-task fallback panel after the full 24-task unseen panel was priced at 147.66 GPU-hours against the 64 GPU-hour branch cap, so full unseen coverage remains missing.

Curriculum by modality. Nine training cells (three modalities under staged, shuffled, and mixed ordering, all under greedy additive best-first and trained fresh from the same base on identical record sets) were evaluated on 27 unseen whole problems. Successes were 24/24/25 (staged/shuffled/mixed) in text-state, 24/27/26 in visual-state, and 25/27/27 in multimodal-state. A sequential-SFT comparator reached 24/27, 25/27, and 24/27 in text, visual, and multimodal cells respectively, the base stayed at 0/27, and random-valid and exact execution saturated at 27/27 in every modality. All four paired interaction intervals include zero: $-0.111 [-0.259, +0.037]$ and $-0.074 [-0.259, +0.074]$ for the visual and multimodal staged-minus-shuffled contrasts against text, and $-0.074 [-0.259, +0.111]$ and $-0.037 [-0.185, +0.111]$ for the corresponding mixed-minus-shuffled contrasts. No material ordering-by-modality interaction emerged. Because both references saturate and the base sits at floor, ceiling and floor effects bound the observable differences.

Transfer. All 12 verified planning adapters plus the base were evaluated zero-shot on frozen subsets of FOLIO (Han et al., 2022) (200 items), GSM8K (Cobbe et al., 2021) (200), and HumanEval (Chen et al., 2021) (164). Table 12 gives the per-adapter accuracies. The paired verdict uses McNemar exact comparisons (McNemar, 1947) of each adapter against the per-example base predictions, Holm-corrected across all 36 comparisons (Holm, 1979). None survives correction: the minimum adjusted p is 0.8156, and the smallest raw p is 0.0227 (FOLIO with the text width-based adapter, at +12 examples). Leakage screening over the frozen 8-gram window finds 7/564 benchmark items sharing only trivial numeric 8-grams with the planning corpus and zero content overlap. The transfer result is a null.

Second backbone. Two reduced-scope cells trained and evaluated InternVL3.5-8B (Wang et al., 2025). Each used a frozen 12-task key-cell panel selected by lowest reference decision count before any model outcome, with one seed-17 training run per modality cell. The earlier cell (protocol v2) ran BFS, and neither arm reached a goal. The pretrained base scored 0/36 at 1 decision per episode (36 total), process SFT scored 0/36 at 1 to 4 decisions (59 total), and the reused BFS controls scored random-valid 54/72 (18/24 per modality, oracle-assisted) and exact 72/72. The replication cell (protocol v3, stress test R3) ran greedy additive best-first with the same backbone, recipe, and panel shape. Process SFT succeeds on 11/12 text-state, 10/12 visual-state, and 10/12 multimodal-state problems (31/36), while the base scores 0/36 with one invalid call per episode. Process-SFT invalid-operation rates are 0.00546, 0.01220, and 0.01258, and decision usage is 183, 164, and 159 of a 390-call allowance (text, visual, multimodal). Random-valid and exact reference both score 24/24 per modality. These control counts hold two entries per panel task, and both entries of a task come from its one reused, hash-pinned greedy baseline control episode per condition and modality. Paired whole-problem contrasts (seed 1729, 10,000 resamples) give SFT minus base $+0.917 [+0.750, +1.000]$ in text-state and $+0.833 [+0.583, +1.000]$ in both visual-state and multimodal-state. SFT minus random-valid is $-0.083 [-0.250, +0.000]$, $-0.167 [-0.417, +0.000]$, and $-0.167 [-0.417, +0.000]$ in the same order. Coverage is complete, with 72/72 model episodes and 72/72

comparators replayed. Learned success under the enumeration contract appears only in the additive cells, where the identity audit shows no decision to make (Section 5.2), and the InternVL v3 cell replicates that contract fluency (31/36 versus 0/36 base, below random-valid 24/24 per modality). The earlier total null on this backbone is consistent with an algorithm (choice-requirement) difference rather than a backbone null. The cross-backbone contrast against the pinned Qwen3-VL process-SFT rate is descriptive rather than paired, because the InternVL panel is the key-cell subset while the Qwen rate is the pinned full-panel aggregate. It is $+0.000 [-0.167, +0.083]$ in text-state, $-0.083 [-0.333, +0.083]$ in visual-state, and $-0.042 [-0.292, +0.125]$ in multimodal-state. InternVL3.5-8B shares the Qwen3-8B LLM family (Yang et al., 2025), so both cells isolate the vision tower, connector, image tokenization, and multimodal recipe, not the LLM backbone. The full-scope v1 second-backbone branch was withdrawn after its terminal VALID_STOP receipt (closed 2026-09-21) and will not be executed, and its gate is retained as history.

N RESULTS DETAIL: FAILURE-MECHANISM CALIBRATION

A frozen CPU-only calibration mined 2,910 unique replay-verified episodes across the five executed model branches and classified all 905 failures with zero unclassified. The protocol was frozen before any outcomes were computed, and the analysis is descriptive. Three arm-level profiles emerge. The base policy fails in 447 of 447 episodes, always on the first decision: 379 emit a malformed output that never reaches the operation checker, and 67 issue rejected BFS frontier operations. Learned arms fail in 407 of 906 episodes, and the dominant learned failure kind is `other_invariant_violation` with 229 occurrences (56.3% of learned failures): parsed, source-correct, applicable operations that the runtime rejects because they violate the declared search invariant. The remaining learned failures are 72 malformed outputs, 56 inapplicable grounded actions, 27 effect failures, 16 schema failures, and smaller kinds. First failures cluster early, at decision indices 0, 1, and 2 with counts 137, 88, and 46. The oracle control fails 51 times, all late and all by expansion-budget exhaustion (buckets 5–8, 9–16, 17–32, and 33+ hold 9, 9, 3, and 30). Table 3 gives the rejection taxonomy.

Every baseline process-SFT episode under BFS or BFWS fails (144/144). The DAgger-iteration BFS arms fail 155/162, while the learned additive best-first families fail 56 of 546 episodes (10.3%). This pattern matches the primary matrix (Section 5.3). Read together with the horizon profile, process supervision teaches the policy to emit well-formed, source-correct operations, but not to maintain invariant-consistent frontier intent over a full episode: the learned policy fails early on invariant grounds, whereas the oracle control, which always emits valid operations, fails only when its expansion budget runs out late in the episode.

O RESULTS DETAIL: OBSERVATION-CORRUPTION CHANNEL TESTS

The observation-corruption suite evaluated frozen checkpoints, with no retraining, on a prospectively frozen set of $k=9$ admitted tasks under four deterministic corruption families applied at observation time. Clean learned success on these tasks is 0.917 (text-state 0.944, visual-state 0.889, multimodal-state 0.889, with 9 tasks $\times$ 2 algorithms per cell). Table 13 gives the paired whole-problem deltas. Text-channel corruption produces the largest drops. Masking or shuffling the text channel lowers success by up to $-0.944 [-1.000, -0.833]$, and the multimodal arm retains at most 0.167 success under shuffled text. Visual-channel corruption is not material. Blanking or degrading the rendered state costs at most $-0.111 [-0.278, +0.000]$, and every visual interval includes zero. The multimodal channel-isolation contrast, paired by task and algorithm ($n=18$), is -0.806 for text-channel corruption versus -0.111 for visual-channel corruption, giving text minus visual -0.694 with 95% interval $[-0.889, -0.472]$, which is material under the frozen rule. Controls behave as the contract predicts: the corrupted base stays at floor (0/144, all first-decision invalid), random-valid succeeds 72/72 on the new frozen seeds, so per the saturation rule no structural-advantage claim is made, and exact execution remains at 18/18.

A decomposition of the 72 published text-corruption episodes (stress test R2) separates two regimes that the success deltas pool together. Under text masking, every episode of the text-state and multimodal-state cells is all-invalid (18/18 in each, 18 invalid operations over 18 decisions). Each emitted operation fails the schema, so the policy never enters the search. This is output-contract

Table 13: Observation corruption on frozen checkpoints ($k=9$ admitted tasks, paired whole-problem deltas against clean runs, strata below 8 descriptive-only under the frozen rule). “Material” is the frozen protocol’s interval-based verdict.

Corruption	Arm	Corrupted	Clean	Δ	95% interval	Material
Text-masked	text	0.000	0.944	-0.944	$[-1.000, -0.833]$	yes
Text-masked	multimodal	0.000	0.889	-0.889	$[-1.000, -0.722]$	yes
Text-shuffled	text	0.000	0.944	-0.944	$[-1.000, -0.833]$	yes
Text-shuffled	multimodal	0.167	0.889	-0.722	$[-0.889, -0.500]$	yes
Visual-blank	visual	0.833	0.889	-0.056	$[-0.167, +0.000]$	no
Visual-blank	multimodal	0.778	0.889	-0.111	$[-0.278, +0.000]$	no
Visual-degraded	visual	0.833	0.889	-0.056	$[-0.167, +0.000]$	no
Visual-degraded	multimodal	0.778	0.889	-0.111	$[-0.278, +0.000]$	no

Table 14: Menu manipulation on the frozen native adapters (both arms, two additive algorithms). Invalid operations are counted over emitted operations. The distractor pick-rate is counted over decisions with injected distractors.

Family	Success	Invalid ops	Distractor picks
Menu permutation	36/36	0/512	n/a
Distractor injection	0/36	36/60	36/60 (0.6)

destruction, not information loss. Copyability-preserving text shuffling behaves differently, because valid operations keep flowing while the search fails. In the text-state cell 16/18 episodes are partial-valid, 2/18 are all-invalid, and none reaches the goal (18 invalid operations over 49 decisions). In the multimodal-state cell 15/18 are partial-valid, none is all-invalid, and 3/18 reach the goal (15 invalid operations over 105 decisions). This partial-valid pattern is the information-loss signature. A new cross cell applies text masking to the frozen visual-state arm, whose nominal state channel is the image, and it scores 0/18 with every episode all-invalid. Masking the text scaffold destroys the output contract even when images are the nominal state channel.

The language-channel claim survives only in its shuffled form. The masked catastrophe in Table 13 is mostly a loss of the copyable output format rather than a loss of state information. Shuffling the text leaves operations copyable and still stops the search, so under the frozen observation contract the textual serialization carries decision-relevant information for these adapters. Removing or degrading the visual channel barely moves success, including on the visual-state arm whose only rendered-semantics channel was corrupted. Sections 5.7 and 5.8 probe the grounded candidate menu directly. This is a channel finding about the trained controller under the declared adapters, not evidence about intrinsic modality value.

P RESULTS DETAIL: MENU-MANIPULATION RECEIPTS

Menu manipulation (stress test R1) probes how the trained policy responds to changes in the grounded menu. It runs on the frozen image-only adapters of both native arms (Section 5.8), with no retraining, under two families. Menu permutation reorders the candidate menu. Distractor injection adds $k=3$ schema-valid but inapplicable actions to it. Table 14 reports the 72 episodes, all independently replayed within the 170/170 stress-evaluation episodes. Section 5.7 states what the two receipts identify.

Q RESULTS DETAIL: IMAGE-ONLY NATIVE ARMS

The native-arms follow-up (contract v2) trained two image-only observation arms. In `visual-nomem-state` (nomem) and `visual-seq-state` (seq), state facts arrive only as rendered images, and seq adds a window of $K=8$ history frames. Neither arm receives textual search memory, g/h /priority scalars, candidate scores, or membership flags. The grounded, unscored candidate menu is still presented. The v1 admission gate measured pretrained-base format compliance and is retained as history. The v2 amendment, frozen before any trained-arm outcome, replaced it with a pre-registered post-training smoke gate, which both arms passed with 62/62 schema-valid grounded calls against a threshold of 0.5. Four adapters (two arms by two additive best-first algorithms) used the frozen recipe with 512 records, 16 updates, and seed 17. All 216/216 new bindings

were recorded and independently replayed, together with 72/72 reused replay-verified comparators. The window spent 3.2086 of 12 GPU-hours.

On the clean panel, both arms nominally succeed on 18/18 with zero invalid operations, against 16/18 for the published frozen `visual-state` arm. The pretrained base scores 0/36 under both new contracts and exact reference scores 18/18. Random-valid is at ceiling on its five frozen seeds, so under the saturation rule no structural-advantage claim is made. Paired whole-problem contrasts ($n=18$, bootstrap seed 61813, 10,000 resamples, 95%) give `nomem` minus `visual-state` $+0.111$ $[0.000, +0.278]$ and `seq` minus `visual-state` $+0.111$ $[0.000, +0.278]$, neither material. The history window changes nothing on these tasks (`seq` minus `nomem` 0.000 $[0.000, 0.000]$). Learned minus pretrained base is $+1.000$ $[+1.000, +1.000]$ for each arm, which is material.

The pre-registered corruption payoff is a null. Under `seq` the images are the only state and history channel, yet blanking every attached image gives a `seq` corrupted-minus-clean contrast of 0.000 $[0.000, 0.000]$, and degrading every image gives 0.000 $[0.000, 0.000]$. The images-only hypothesis is therefore answered in the negative on this panel. As a post-hoc interpretation, the grounded unscored candidate menu is the remaining channel that could carry the signal, since neither the rendered scenes nor the removed scalar scaffold is necessary once the menu is present. On these additive cells the identity audit (Section 5.2) makes any listed candidate a correct emission, so the null is equally consistent with a policy that uses no state information.

The degraded-strain test (stress test R4) evaluated the same frozen native adapters on the two generalization families where the full-scaffold adapters had degraded. The learned adapters succeed on 10/10 shifted-init and 10/10 scale-up episodes for both arms, and the pretrained base scores 0/20 on each strain (0/40 in total). The published full-scaffold rates on the same variants are 19/30 for shifted-init and 26/30 for scale-up (Appendix M). The two rates come from different episode sets and are not paired. The reduced contract sits at ceiling where the full scaffold degraded. A post-hoc, unpaired reading, with 10 visual-only episodes per arm against 30 full-scaffold episodes pooled over three modalities, suggests that the scaffold’s scalar and bookkeeping text carried no useful signal on these variants.

R RESULTS DETAIL: CHOICE-FRONTIER AND COMPARATOR-ZOO RECEIPTS

The training corpus was derived on CPU by replaying the frozen exact traces through the new contract and verified expansion-for-expansion against them (58 episodes bit-exact). It covers 29 training tasks and holds 1,046 records (512 training plus 11 diagnostic per algorithm) with 3,544 scenes. Every corpus episode carries at least one decision whose menu offers two or more states (up to 315 such decisions in one episode, menus of up to 222 states). Two adapters, one per additive algorithm, were trained with the frozen recipe at seed 17 (16 updates, 512 records each), and the smoke gate accepted 40/40 model calls. All 144/144 panel episodes were independently replayed with complete coverage. The window spent 1.069 of 12 GPU-hours (training 0.81, smoke 0.052, model evaluation 0.208), and corpus and controls ran on CPU.

The control expansion sequences coincide across the two algorithms on every panel task. Post-hoc task-clustered intervals with the same seed and resample count are $[0.000, +0.100]$ for learned minus random-valid and $[-0.889, -0.333]$ for random-valid minus exact reference.

Random-valid is the zoo floor and exact reference the imitation target. Random-valid and the three selectors also ran at $3\times$ and $4\times$ caps, outside the primary metric. Because the CPU selectors operate on the additive frontier, novelty-first selection does not reproduce BFWs pruning. Selector episodes replay the frozen model input, randomized menu, actions, admission, and budget, but do not generate or replay image view bindings. All 432/432 zoo bindings were observed and independently replayed, with 342 new CPU episodes, 90 original $2\times$ random-valid controls reused, and none missing. The nine unit-cost optimal costs C^* come from one offline planner computation per task with replay-verified plans: storage 3; elevators, ferry, and 15puzzle-compact 4; depot, hanoi, and visitall 5; blocksworld and 15puzzle-expanded 6.

The selectors are `bfs-order` (minimum frontier g , then earliest serial), novelty-first selection (lowest width-1 then width-2 novelty over positive PDDL atoms), and `worst-first` as a negative control. The 144 stored choice-frontier episodes enter the metric set post-hoc. On the primary M1 metric the learned adapter scores .1042, between `bfs-order` at .0694 and random-valid at .1389,

Table 15: Prospective training-seed replication for the headline process-SFT cell (greedy best-first $\times$ multimodal-state, 24-problem unseen panel). Seeds 17/29/71 frozen pre-launch. 534 new episodes independently replayed. Aggregation descriptive ($n=3$).

Measure	Seed 17	Seed 29	Seed 71
Success /24	21/24	21/24	22/24
SFT – base	+0.875	+0.875	+0.917
SFT – random-valid	−0.125	−0.125	−0.083

and below novelty-first selection at .1250. The fifteen-test Holm family pairs the learned adapter with each of five comparators on three metrics.

Exploratory M4 on the stored episodes shows the learned adapter selecting the teacher’s heap-head state on 23/132 multi-option decisions (17.4%) against an on-policy chance rate of 26.9%, and emitting the last menu label on 72/132 (54.5%). The post-hoc choice-frontier diagnostic records a first-label rate of 2.3% and a teacher-corpus last-label rate of 12.0% (chance 10.4%, 978 records). Menus are permuted per decision with a seed, so a last-label habit acts operationally as a random picker. M5 restricts to tasks where random-valid solves at a $2\times$ rate strictly between 0 and 1. That subset holds one task (storage-compact-919000, random-valid rate .2), so M5 is descriptive only. On it the learned adapter’s M1 AUC is .3125, against .075 for random-valid and .625 for exact reference.

The enumeration-contract identity audit records its structural basis in the artifact. Each decision submits one remaining candidate, `finish_expansion()` requires the complete candidate set, and heap serials follow the deterministic sorted candidate order at `start_expansion`, so frontier evolution is invariant to submission order.

The choice-frontier contrasts use percentile 95% intervals with the frozen bootstrap resampling cells. Random-valid’s seed frequency on storage is 0.2. The final audit returns `CHOICE_SENSITIVE` with `ok` over 144 replayed episodes.

S RESULTS DETAIL: SEED REPLICATION AND FOLLOW-UP LIMITS

The prospective training-seed replication (seeds 17/29/71 frozen pre-launch, 534 new episodes independently replayed) executed on two cells (Table 15). On the headline process-SFT cell (greedy best-first $\times$ multimodal-state, 24-problem unseen panel), per-seed success is 21/24, 21/24, 22/24 (mean 0.889, range [0.875, 0.917]). The contrast against the pretrained base is positive at every seed (+0.875/+0.875/+0.917, no sign flip). The contrast against random-valid stays negative (−0.125/−0.125/−0.083). On the DAgger iteration-1 cell set (BFS $\times$ three modalities, DAgger and continued-SFT arms), every arm and seed scores 0/72 unseen, with paired contrast +0.000 [+0.000, +0.000] at each seed. The published iteration-2 verdict (1/72 vs 1/72, seed 17) is unchanged. Both cells replicate across all three seeds. Cell A covers one of the three pooled headline modalities. The remaining two modalities and all other cells use seed 17 only. Cross-seed aggregation is descriptive ($n=3$). The image-only-arm proposal has since executed under the native-arms v2 amendment (Section 5.8), and its v1 gate remains as history.

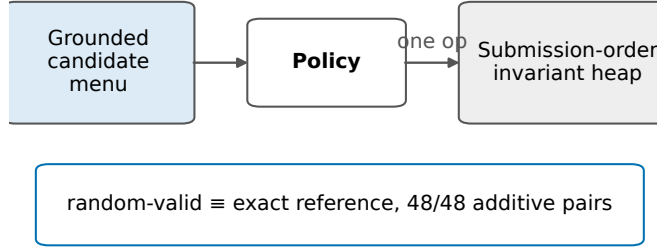
The follow-up windows share this development-stage status, and we list their limits separately. The native-arms window evaluates an 18-unit clean panel with one training seed per cell. The R1-R4 follow-up window reuses the frozen native-arms adapters for R1, R2, and R4, and trains one seed-17 run per cell for R3. The choice-frontier window trains one seed per algorithm on a 9-task panel whose discriminative subset holds one task. The comparator zoo is a CPU-only exploratory analysis. None of these windows accessed the held-out manifest. The shared nine-task panel draws from 8 domains with 7 compact and 2 expanded variants.

Final primary evaluation prerequisites. No authorized held-out evaluation exists. Freezing one requires recorded decisions on the following prerequisites. (1) The condition-level data source and Search-Trace Segment construction with per-algorithm allocation. (2) Compute parity and checkpoint-selection rules. (3) The episode-budget unit, the whole-instance split procedure, and the fixed Search Memory capacity. (4) The estimand, the uncertainty interval or test, the success threshold, the invariant-compliance rule, and the validity and invalid-operation denominators for every cell. Until these choices are recorded, the held-out evaluation cannot be frozen and no final comparative efficacy finding exists.

Table 16: The two decision contracts. Rows compare what the policy emits, who owns the frontier, what the observation carries, how the budget binds, and the role of the identity audit. Results under the two contracts are not comparable as performance rows.

	Enumeration contract	Choice-frontier contract
Policy emits	One Typed Search Operation per step, taken from a grounded, unscored candidate action menu	One opaque label from a menu of frontier states, naming the state to expand next
Frontier	Kept in textual Search Memory with visited and score state, advanced by submitted operations	Owned by the runtime. Menu labels are re-assigned per decision by a seeded permutation
Observation	Set by the Modality Observation adapter	Visual only, with task pages, the initial scene, one unlabelled 128px scene per frontier state, and goal pages
Budget	Learned arms capped at $2\times$ the exact-reference decision count, base at one call	Decisions and expansions capped at $2\times$ the additive exact-reference expansion count, goal selection uncharged
Identity audit	Diagnostic run on the expanded-baseline additive cells	Pre-registered pass condition over 18 control pairs

(a) Enumeration contract



(b) Choice-frontier contract

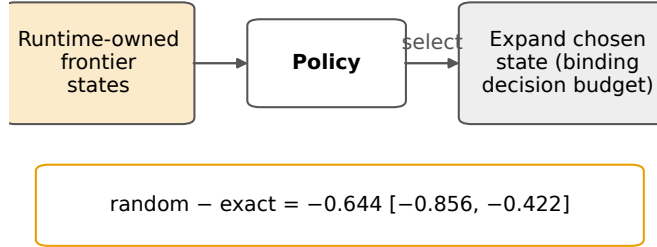


Figure 2: Two decision contracts on the same additive task family. Under the enumeration contract each decision submits one remaining candidate and the heap ignores submission order, so random-valid and the exact reference are decision- and expansion-identical (48/48 pairs). Under the choice-frontier contract the policy selects which frontier state to expand under a binding budget, and random-valid separates from exact (-0.644 , 95% CI $[-0.856, -0.422]$).

T DECISION CONTRACTS

Table 16 compares the enumeration and choice-frontier contracts defined in Section 4.

U USE OF LLM STATEMENT

An LLM was used only for polishing language; the scientific content is entirely original.